

UGA Mock-3

Time Allowed: 2 Hours

Maximum Marks: 120

Instructions: Each question has exactly one correct option.

1. Let $P(x)$ be a polynomial of degree 10 such that $P(k) = \frac{1}{k+1}$ for $k = 0, 1, 2, \dots, 10$. Find the value of $P(11)$.
(A) $1/12$ (B) 0 (C) $1/11$ (D) $1/6$
2. What is the highest power of 2 that strictly divides the number $3^{1024} - 1$?
(A) 10 (B) 11 (C) 13 (D) 12
3. Let $S = \{1, 2, \dots, 15\}$. Find the number of subsets of S such that the sum of their elements is a multiple of 3. (Assume the sum of the empty set is 0).
(A) 10944 (B) 10923 (C) 10924 (D) 10925
4. Let $f(x) = x^3 - 3x + 1$. Determine the total number of distinct real roots of the equation $f(f(x)) = 0$.
(A) 5 (B) 7 (C) 9 (D) 11
5. Evaluate
$$\lim_{n \rightarrow \infty} n \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} - \ln 2 \right)$$

(A) $-1/4$ (B) $1/4$ (C) $-1/2$ (D) $1/2$
6. Let A and B be $n \times n$ matrices with real entries such that $AB - BA = A$. What is the trace of the matrix A^{2026} ?
(A) n (B) 2026 (C) 0 (D) 1
7. Let P be a point located inside a regular tetrahedron $ABCD$ of edge length 1. The sum of the squared distances $PA^2 + PB^2 + PC^2 + PD^2$ attains its absolute minimum value. What is this minimum value?
(A) 1 (B) $3/2$ (C) 2 (D) $4/3$

8. Let the roots of the equation $z^3 - 6z^2 + 12z - 16 = 0$ represent the vertices of a triangle in the complex plane. The area of this triangle is

(A) $2\sqrt{3}$ (B) $4\sqrt{3}$ (C) $6\sqrt{3}$ (D) $3\sqrt{3}$

9. Let $\sigma(n)$ denote the sum of all positive divisors of n . Find the number of integers n in the set $\{1, 2, \dots, 100\}$ such that $\sigma(n)$ is an odd integer.

(A) 10 (B) 15 (C) 17 (D) 20

10. Find the value of

$$\int_0^1 \frac{\ln(1+x)}{1+x^2} dx$$

(A) $\frac{\pi}{4} \ln 2$ (B) $\frac{\pi}{8} \ln 2$ (C) $\frac{\pi}{2} \ln 2$ (D) $\frac{\pi}{4} \ln 3$

11. What is the remainder when the number $10^{10^{10}}$ is divided by 7?

(A) 1 (B) 2 (C) 3 (D) 4

12. Let α, β, γ be the three roots of the cubic equation $x^3 - x - 1 = 0$. Evaluate the expression $\frac{1-\alpha}{1+\alpha} + \frac{1-\beta}{1+\beta} + \frac{1-\gamma}{1+\gamma}$.

(A) 1 (B) -1 (C) 3 (D) 7

13. Find the number of ordered pairs of positive integers (x, y) that satisfy the equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{2026}$.

(A) 3 (B) 5 (C) 7 (D) 9

14. Let $f(x) = \lim_{n \rightarrow \infty} \frac{x^{2n} - 1}{x^{2n} + 1}$ for real x . At how many distinct points in its domain is the function $f(x)$ discontinuous?

(A) 0 (B) 1 (C) 2 (D) 3

15. Find the number of ordered pairs of sets (A, B) such that $A \cup B = \{1, 2, \dots, n\}$ and the intersection $A \cap B$ contains exactly one element.

(A) $n \cdot 3^{n-1}$ (B) $n \cdot 2^{n-1}$ (C) $2^n - n$ (D) $n \cdot 2^n$

16. How many binary sequences of length 14 contain no two consecutive zeros?

(A) 233 (B) 987 (C) 610 (D) 377

17. Find the value of

$$\int_0^\infty \frac{\ln x}{1+x^2} dx$$

(A) $\pi/2$ (B) $\pi/4$ (C) 0 (D) ∞

18. What is the remainder when $\sum_{k=1}^{2026} k^k$ is divided by 4?

- (A) 0 (B) 2 (C) 1 (D) 3

19. Find the total area enclosed by the curve given by the equation $(x^2 + y^2)^2 = x^2 - y^2$.

- (A) $1/2$ (B) 1 (C) $\pi/2$ (D) $\pi/4$

20. Evaluate

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

- (A) 4 (B) 6 (C) 8 (D) 12

21. Let x_1, x_2, x_3 be the three complex roots of the polynomial equation $x^3 - 3x^2 + 1 = 0$. Evaluate $(x_1^5 + x_2^5 + x_3^5)$.

- (A) 198 (B) 201 (C) 195 (D) 204

22. Find the value of

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

- (A) $\pi^2/2$ (B) $\pi/4$ (C) $\pi/2$ (D) $\pi^2/4$

23. Let z_1, z_2, z_3, z_4, z_5 be the five complex roots of $z^5 - 1 = 0$, where $z_5 = 1$. What is the value of $\prod_{k=1}^4 (3 - z_k)$?

- (A) 121 (B) 122 (C) 243 (D) 242

24. Let $f(x)$ be a strictly increasing convex function defined on the interval $[0, 1]$ such that $f(0) = 0$ and $f(1) = 1$. Evaluate $\lim_{n \rightarrow \infty} n \int_0^1 x^n f(x) dx$.

- (A) 0 (B) $1/2$ (C) 1 (D) e

25. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying $f(x+y) + f(x-y) = 2f(x) \cos y$, for all $x, y \in \mathbb{R}$. If $f(0) = 1$ and $f(\pi/2) = 1$, find the value of $f(\pi)$.

- (A) 1 (B) 0 (C) -1 (D) π

26. Let S be the sum of the coefficients of the even powers of x in the polynomial expansion of $P(x) = (x^2 - 2x + 2)^{2026}$. Find S .

- (A) $\frac{5^{2026} + 1}{2}$ (C) 5^{2026}
(B) $\frac{5^{2026} - 1}{2}$ (D) 1

- 27.** A sequence is defined recursively by $a_1 = 3$ and $a_{n+1} = a_n^2 - 2$ for all integers $n \geq 1$. Find the remainder when a_{2026} is divided by 13.

(A) 3 (B) 7 (C) 8 (D) 10

- 28.** Evaluate the sum

$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)(k+2)}$$

.

(A) $1/2$ (B) $1/4$ (C) $1/6$ (D) $1/8$

- 29.** For real numbers x, y , the absolute minimum possible value of the expression

$$\sqrt{x^2 + (y-1)^2} + \sqrt{(x-1)^2 + y^2}$$

is

(A) 1 (B) 2 (C) $2\sqrt{2}$ (D) $\sqrt{2}$

- 30.** Consider the recursively defined sequence $a_1 = 1$, and $a_{n+1} = a_n + \frac{1}{a_n}$ for $n \geq 1$. Evaluate

$$\lim_{n \rightarrow \infty} \frac{a_n}{\sqrt{n}}$$

(A) 1 (B) 2 (C) ∞ (D) $\sqrt{2}$

Answer Key

1. D	7. B	13. D	19. B	25. C
2. D	8. D	14. C	20. B	26. A
3. A	9. C	15. B	21. A	27. D
4. B	10. B	16. B	22. D	28. B
5. A	11. D	17. C	23. A	29. D
6. C	12. A	18. C	24. C	30. D

UGA Mock-3 Solutions

Question 1 - Solution

Given $P(k) = \frac{1}{k+1}$ for $k = 0, 1, 2, \dots, 10$. Let us define a new polynomial $Q(x) = (x + 1)P(x) - 1$. Since $P(x)$ has a degree of 10, $Q(x)$ must be a polynomial of degree 11. The roots of $Q(x) = 0$ are exactly $x = 0, 1, 2, \dots, 10$. Thus, we can express $Q(x)$ as:

$$Q(x) = c \cdot x(x-1)(x-2) \cdots (x-10)$$

where c is a constant. We can find c by substituting $x = -1$:

$$Q(-1) = (-1+1)P(-1) - 1 = -1$$

Also, using the factored form:

$$Q(-1) = c(-1)(-2)(-3) \cdots (-11) = -c \cdot 11!$$

Equating the two expressions gives:

$$-c \cdot 11! = -1 \implies c = \frac{1}{11!}$$

We need to find $P(11)$. We substitute $x = 11$ into our expression for $Q(x)$:

$$Q(11) = c \cdot 11 \cdot 10 \cdot 9 \cdots 1 \implies Q(11) = c \cdot 11! \implies Q(11) = \frac{1}{11!} \cdot 11! = 1$$

Using the definition $Q(x) = (x+1)P(x) - 1$, for $x = 11$:

$$(11+1)P(11) - 1 = 1 \implies 12P(11) = 2 \implies P(11) = \frac{1}{6}$$

Hence, option (D) is correct.

Question 2 - Solution

We need the highest power of 2 dividing $3^{1024} - 1$. We use the Lifting the Exponent Lemma (LTE) for $p = 2$. For any odd integer a and an integer n that is even, $v_2(a^n - 1) = v_2(a - 1) + v_2(a + 1) + v_2(n) - 1$. Here, $a = 3$ and $n = 1024 = 2^{10}$.

$$\begin{aligned} v_2(3^{1024} - 1) &= v_2(3 - 1) + v_2(3 + 1) + v_2(2^{10}) - 1 \\ \implies v_2(3^{1024} - 1) &= v_2(2) + v_2(4) + 10 - 1 \implies 1 + 2 + 10 - 1 = 12 \end{aligned}$$

Hence, option (D) is correct.

Question 3 - Solution

We want subsets of $S = \{1, 2, \dots, 15\}$ where the sum is a multiple of 3. Let $w = e^{2\pi i/3}$. By the roots of unity filter, the number of subsets N_0 is given by:

$$N_0 = \frac{1}{3} \sum_{j=0}^2 \prod_{k=1}^{15} (1 + w^{jk})$$

For $j = 0$, the term is $2^{15} = 32768$. For $j = 1$ and $j = 2$, the products are equal to $\prod_{k=1}^{15} (1 + w^k)$. Since k covers each residue modulo 3 exactly 5 times, we group terms as $((1 + w)(1 + w^2)(1 + w^3))^5$. Since $1 + w = -w^2$, $1 + w^2 = -w$, and $1 + w^3 = 2$, their product is $(-w^2)(-w)(2) = 2$. Thus, the product for $j = 1$ and $j = 2$ is $2^5 = 32$.

$$N_0 = \frac{1}{3}(32768 + 32 + 32) \implies N_0 = 10944$$

Hence, option (A) is correct.

Question 4 - Solution

We are given $f(x) = x^3 - 3x + 1$ and must find the number of distinct real roots of $f(f(x)) = 0$. First, find the roots of $f(y) = 0$. Observing the values, $f(-2) = -1$, $f(-1) = 3$, $f(0) = 1$, $f(1) = -1$, and $f(2) = 3$. Since the polynomial changes signs, it has three real roots: $y_1 \in (-2, -1)$, $y_2 \in (0, 1)$, and $y_3 \in (1, 2)$. Now we must solve $f(x) = y_1$, $f(x) = y_2$, and $f(x) = y_3$. The local maximum of $f(x)$ is at $x = -1 \implies f(-1) = 3$ and the local minimum is at $x = 1 \implies f(1) = -1$. For $y_1 \in (-2, -1)$, y_1 is below the local minimum $\implies f(x) = y_1$ has 1 real root. For $y_2 \in (0, 1)$, y_2 is between the local minimum and maximum $\implies f(x) = y_2$ has 3 real roots. For $y_3 \in (1, 2)$, y_3 is also between the local minimum and maximum $\implies f(x) = y_3$ gives 3 real roots. Total distinct real roots $= 1 + 3 + 3 = 7$.

Hence, option (B) is correct.

Question 5 - Solution

We evaluate $\lim_{n \rightarrow \infty} n \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} - \ln 2 \right)$. We substitute the asymptotic expansion of the harmonic numbers $H_n \approx \ln n + \gamma + \frac{1}{2n} - \frac{1}{12n^2}$.

$$\begin{aligned} \sum_{k=1}^n \frac{1}{n+k} &= H_{2n} - H_n = \left(\ln(2n) + \gamma + \frac{1}{4n} - \frac{1}{48n^2} \right) - \left(\ln n + \gamma + \frac{1}{2n} - \frac{1}{12n^2} \right) \\ &\implies H_{2n} - H_n = \ln 2 - \frac{1}{4n} + O\left(\frac{1}{n^2}\right) \end{aligned}$$

The final answer is

$$\lim_{n \rightarrow \infty} n \left(-\frac{1}{4n} \right) = -\frac{1}{4}.$$

Hence, option (A) is correct.

Question 6 - Solution

For $n \times n$ matrices satisfying $AB - BA = A$, we must find the trace of A^{2026} . Taking the trace of both sides of $A = AB - BA$, we get $\text{Tr}(A) = \text{Tr}(AB) - \text{Tr}(BA) \implies \text{Tr}(A) = 0$. We multiply A on the left iteratively: $A^k = A^k B - A^{k-1} BA$. It is a standard result that, for such matrices A , and B , we have $A^k B - BA^k = kA^k$. Taking the trace gives:

$$k\text{Tr}(A^k) = \text{Tr}(A^k B) - \text{Tr}(BA^k) = 0 \implies \text{Tr}(A^k) = 0$$

Thus, for $k = 2026$, $\text{Tr}(A^{2026}) = 0$.

Hence, option (C) is correct.

Question 7 - Solution

Let G be the centroid of the regular tetrahedron $ABCD$ with edge 1. Using the scalar product or Apollonius's generalized theorem, for any point P :

$$PA^2 + PB^2 + PC^2 + PD^2 = 4PG^2 + GA^2 + GB^2 + GC^2 + GD^2$$

To minimize this sum, P must be placed exactly at $G \implies PG = 0$, giving a minimum value of $4GA^2$. For a regular tetrahedron of edge length $a = 1$, the distance from the centroid to any vertex is $GA = \frac{\sqrt{6}}{4}$. Squaring this gives $GA^2 = \frac{6}{16} \implies GA^2 = \frac{3}{8}$. The minimum sum is $4 \times \frac{3}{8} = \frac{3}{2}$.

Hence, option (B) is correct.

Question 8 - Solution

The roots are of the equation $z^3 - 6z^2 + 12z - 16 = 0$. This can be rewritten by noticing it resembles $(z - 2)^3$:

$$(z - 2)^3 = z^3 - 6z^2 + 12z - 8$$

Thus, the equation is:

$$(z - 2)^3 - 8 = 0 \implies (z - 2)^3 = 8 \implies z - 2 = 2e^{i2\pi k/3}$$

This means the roots form an equilateral triangle in the complex plane, centered at $z = 2$, with a circumradius $R = 2$. The side length s of an equilateral triangle inscribed in a circle of radius R is $R\sqrt{3} \implies s = 2\sqrt{3}$. The area is:

$$\text{Area} = \frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}(12) \implies \text{Area} = 3\sqrt{3}$$

Hence, option (D) is correct.

Question 9 - Solution

We are searching for integers n up to 100 where $\sigma(n)$ is an odd integer. The sum of divisors $\sigma(n)$ is odd if and only if n is either a perfect square or twice a perfect square. Perfect squares from 1 to 100: $1, 4, 9, 16, 25, 36, 49, 64, 81, 100 \implies 10$ numbers. Twice a perfect square up to 100: $2(1^2) = 2, 2(2^2) = 8, 2(3^2) = 18, 2(4^2) = 32, 2(5^2) = 50, 2(6^2) = 72, 2(7^2) = 98 \implies 7$ numbers. Total valid integers $= 10 + 7 = 17$.

Hence, option (C) is correct.

Question 10 - Solution

Evaluate $I = \int_0^1 \frac{\ln(1+x)}{1+x^2} dx$. Using the substitution $x = \tan \theta \implies dx = \sec^2 \theta d\theta$. The limits change to 0 and $\pi/4$.

$$I = \int_0^{\pi/4} \ln(1 + \tan \theta) d\theta$$

Applying the property $\int_0^a f(x) dx = \int_0^a f(a-x) dx$:

$$\begin{aligned}\Rightarrow 1 + \tan(\pi/4 - \theta) &= 1 + \frac{1 - \tan \theta}{1 + \tan \theta} = \frac{2}{1 + \tan \theta} \\ \Rightarrow I &= \int_0^{\pi/4} \ln \left(\frac{2}{1 + \tan \theta} \right) d\theta = \int_0^{\pi/4} (\ln 2 - \ln(1 + \tan \theta)) d\theta \\ \Rightarrow I &= \frac{\pi}{4} \ln 2 - I \implies 2I = \frac{\pi}{4} \ln 2 \implies I = \frac{\pi}{8} \ln 2\end{aligned}$$

Hence, option (B) is correct.

Question 11 - Solution

We need $10^{10^{10}} \pmod{7}$. Since $10 \equiv 3 \pmod{7}$, this is $3^{10^{10}} \pmod{7}$. By Fermat's Little Theorem, $3^6 \equiv 1 \pmod{7}$, so we need to find $10^{10} \pmod{6}$. Since $10 \equiv 4 \pmod{6}$, and any power of 4 modulo 6 is 4 ($4^2 = 16 \equiv 4$), we have $10^{10} \equiv 4 \pmod{6}$. Therefore:

$$3^{10^{10}} \equiv 3^4 = 81 \equiv 4 \pmod{7}$$

The remainder is 4.

Hence, option (D) is correct.

Question 12 - Solution

Given roots α, β, γ of $x^3 - x - 1 = 0$, evaluate the expression with $y = \frac{1-x}{1+x}$. Rearranging for x gives $x = \frac{1-y}{1+y}$. We substitute this into the polynomial:

$$\left(\frac{1-y}{1+y} \right)^3 - \frac{1-y}{1+y} - 1 = 0$$

Multiplying by $(1+y)^3$:

$$(1-y)^3 - (1-y)(1+y)^2 - (1+y)^3 = 0 \implies y^3 - y^2 + 7y + 1 = 0$$

The roots of this new cubic are y_1, y_2, y_3 . The sum of the roots is given by:

$$-\frac{\text{coefficient of } y^2}{\text{coefficient of } y^3} = -\frac{-1}{1} = 1$$

Hence, option (A) is correct.

Question 13 - Solution

We want positive integer pairs (x, y) such that $\frac{1}{x} + \frac{1}{y} = \frac{1}{2026}$. Rearranging this gives:

$$(x - 2026)(y - 2026) = 2026^2$$

Let $x - 2026 = a$ and $y - 2026 = b$. Since $x, y > 0$, we need the number of positive divisors of 2026^2 . Factoring $2026 = 2 \times 1013$. 1013 is a prime number. Thus, $2026^2 = 2^2 \times 1013^2$. The number of positive divisors is $(2 + 1)(2 + 1) \implies 9$. This means there are 9 integer pairs $(a, b) \implies 9$ ordered pairs (x, y) .

Hence, option (D) is correct.

Question 14 - Solution

Let $f(x) = \lim_{n \rightarrow \infty} \frac{x^{2n} - 1}{x^{2n} + 1}$. We evaluate this piecewise: If $|x| < 1$, $x^{2n} \rightarrow 0 \implies f(x) = -1/1 = -1$. If $|x| > 1$, $x^{2n} \rightarrow \infty \implies f(x) = 1$. If $|x| = 1$, $x^{2n} = 1 \implies f(x) = 0/2 = 0$. The function jumps at $x = 1$ and $x = -1$. Therefore, it is discontinuous at 2 distinct points.

Hence, option (C) is correct.

Question 15 - Solution

Find pairs of sets (A, B) such that $A \cup B = \{1, 2, \dots, n\}$ and $A \cap B$ contains exactly one element. There are exactly n ways to choose the single element that belongs to $A \cap B$. For the remaining $n - 1$ elements in the union, each must belong to either strictly A or strictly B (but not both). This gives 2 choices for each of the $n - 1$ elements $\implies 2^{n-1}$ combinations. Total combinations = $n \cdot 2^{n-1}$.

Hence, option (B) is correct.

Question 16 - Solution

We are counting binary sequences of length 14 without consecutive zeros. Let a_n be the number of valid sequences of length n . Appending a 1 to a valid $n - 1$ sequence keeps it valid (a_{n-1} ways). Appending a 0 requires the previous digit to be 1, which effectively adds "10" to a valid $n - 2$ sequence (a_{n-2} ways). This gives the relation:

$$a_n = a_{n-1} + a_{n-2}$$

For $n = 1$, sequences are 0, 1 $\implies a_1 = 2$. For $n = 2$, sequences are 01, 10, 11 $\implies a_2 = 3$. This tracks with Fibonacci numbers where $a_n = F_{n+2}$ (starting with $F_1 = F_2 = 1$). For $n = 14$, we need F_{16} . Expanding Fibonacci gives $F_{16} = 987$.

Hence, option (B) is correct.

Question 17 - Solution

To evaluate $I = \int_0^\infty \frac{\ln x}{1+x^2} dx$, apply the substitution $x = 1/t \implies dx = -1/t^2 dt$. The limits flip from ∞ to 0.

$$I = \int_\infty^0 \frac{\ln(1/t)}{1 + (1/t)^2} \left(-\frac{1}{t^2}\right) dt = \int_0^\infty \frac{-\ln t}{1 + t^2} dt = -I$$

Since $I = -I \implies 2I = 0 \implies I = 0$.

Hence, option (C) is correct.

Question 18 - Solution

Find the remainder when $\sum_{k=1}^{2026} k^k$ is divided by 4. Evaluate $k^k \pmod{4}$ based on congruence classes modulo 4: If $k \equiv 0 \pmod{4} \implies k^k \equiv 0$ for $k \geq 4$. If $k \equiv 1 \pmod{4} \implies k^k \equiv 1^1 = 1$. If $k \equiv 2 \pmod{4} \implies k^k \equiv 2^k \equiv 0$ for $k \geq 2$. If $k \equiv 3 \pmod{4} \implies k^k \equiv (-1)^{\text{odd}} \equiv 3$. Every consecutive block of 4 sums to $0 + 1 + 0 + 3 = 4 \implies 0 \pmod{4}$ (the first block is $1^1 + 2^2 + 3^3 + 4^4 = 288 \equiv 0$). We have $\lfloor 2026/4 \rfloor = 506$ blocks, leaving 2 terms: $k = 2025, 2026$. $2025 \equiv 1 \implies 1$, and $2026 \equiv 2 \implies 0$. Summing the remainder: $1 + 0 = 1$.

Hence, option (C) is correct.

Question 19 - Solution

To find the area enclosed by $(x^2 + y^2)^2 = x^2 - y^2$, convert to polar coordinates where $x^2 + y^2 = r^2$ and $x^2 - y^2 = r^2 \cos 2\theta$:

$$r^4 = r^2 \cos 2\theta \implies r^2 = \cos 2\theta$$

This figure is known as lemniscate. The total area is obtained by integrating r^2 over the two loops. One loop spans from $-\pi/4$ to $\pi/4$.

$$\text{Area} = 2 \int_{-\pi/4}^{\pi/4} \frac{1}{2} r^2 d\theta = \int_{-\pi/4}^{\pi/4} \cos 2\theta d\theta = \left[\frac{1}{2} \sin 2\theta \right]_{-\pi/4}^{\pi/4} = \frac{1}{2} (1 - (-1)) = 1$$

Hence, option (B) is correct.

Question 20 - Solution

Evaluate $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$. Using the generating function $S(x) = \sum_{n=1}^{\infty} x^n = \frac{x}{1-x}$: Applying the operator $x \frac{d}{dx}$ twice:

$$x \frac{d}{dx} \left(x \frac{d}{dx} \frac{x}{1-x} \right) = \frac{x(1+x)}{(1-x)^3} = \sum_{n=1}^{\infty} n^2 x^n$$

Substitute $x = 1/2$:

$$S(1/2) = \frac{(1/2)(1+1/2)}{(1-1/2)^3} = \frac{3/4}{1/8} = 6$$

Hence, option (B) is correct.

Question 21 - Solution

Let x_1, x_2, x_3 be roots of $x^3 - 3x^2 + 1 = 0$. Evaluate the sum of 5th powers S_5 . The characteristic equation gives $x^3 = 3x^2 - 1 \implies S_n = 3S_{n-1} - S_{n-3}$. $S_0 = 3$. $S_1 = 3$. $S_2 = (\sum x_i)^2 - 2 \sum x_i x_j = 3^2 - 2(0) = 9$. $S_3 = 3S_2 - S_0 = 3(9) - 3 = 24$. $S_4 = 3S_3 - S_1 = 3(24) - 3 = 69$. $S_5 = 3S_4 - S_2 = 3(69) - 9 = 207 - 9 = 198$.

Hence, option (A) is correct.

Question 22 - Solution

Evaluate $I = \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$. Apply $\int_0^\pi f(x) dx = \int_0^\pi f(\pi - x) dx$:

$$I = \int_0^\pi \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx \implies I = \pi \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx - I \implies 2I = \pi \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx$$

Let $u = \cos x \implies du = -\sin x dx$, with limits 1 to -1.

$$2I = \pi \int_{-1}^1 \frac{1}{1 + u^2} du \implies \pi [\arctan u]_{-1}^1 = \pi \left(\frac{\pi}{4} - \left(-\frac{\pi}{4} \right) \right) = \frac{\pi^2}{2} \implies I = \frac{\pi^2}{4}$$

Hence, option (D) is correct.

Question 23 - Solution

The complex roots z_1, z_2, z_3, z_4 are the non-trivial roots of $z^5 - 1 = 0$. We evaluate $\prod_{k=1}^4 (3 - z_k)$. The polynomial factorization gives:

$$z^5 - 1 = (z - 1)(z^4 + z^3 + z^2 + z + 1)$$

The roots of the second factor are z_1, z_2, z_3, z_4 . Thus, the polynomial evaluates as:

$$\prod_{k=1}^4 (z - z_k) = z^4 + z^3 + z^2 + z + 1$$

Substituting $z = 3$:

$$3^4 + 3^3 + 3^2 + 3 + 1 = 81 + 27 + 9 + 3 + 1 = 121$$

Hence, option (A) is correct.

Question 24 - Solution

Let $f(x)$ be strictly increasing convex on $[0, 1]$ with $f(0) = 0, f(1) = 1$. We evaluate $\lim_{n \rightarrow \infty} n \int_0^1 x^n f(x) dx$. Integrating by parts with $u = f(x)$ and $dv = nx^n dx \implies v = \frac{n}{n+1} x^{n+1}$.

$$I_n = \left[\frac{n}{n+1} x^{n+1} f(x) \right]_0^1 - \int_0^1 \frac{n}{n+1} x^{n+1} f'(x) dx = \frac{n}{n+1} f(1) - \frac{n}{n+1} \int_0^1 x^{n+1} f'(x) dx$$

As $n \rightarrow \infty$, $\frac{n}{n+1} \rightarrow 1$. Because $f'(x)$ is bounded (or at worst continuous over the compact interval excluding endpoints where properties converge), $\int_0^1 x^{n+1} f'(x) dx \rightarrow 0$. The limit evaluates to:

$$1 \cdot f(1) - 0 = 1.$$

Hence, option (C) is correct.

Question 25 - Solution

Given $f(x+y) + f(x-y) = 2f(x)\cos y$. Substitute $y = \pi/2$, which makes $\cos(\pi/2) = 0$:

$$f(x + \pi/2) + f(x - \pi/2) = 0 \implies f(x + \pi/2) = -f(x - \pi/2)$$

Letting $t = x - \pi/2$, we see:

$$f(t + \pi/2 + \pi/2) = -f(t) \implies f(t + \pi) = -f(t)$$

Substitute $t = 0 \implies f(\pi) = -f(0)$. We are given $f(0) = 1 \implies f(\pi) = -1$.

Hence, option (C) is correct.

Question 26 - Solution

Find S , the sum of coefficients of the even powers in $P(x) = (x^2 - 2x + 2)^{2026}$. For any polynomial, the sum of even-powered coefficients is $\frac{P(1)+P(-1)}{2}$.

$$P(1) = (1^2 - 2(1) + 2)^{2026} = 1^{2026} = 1,$$

and

$$\begin{aligned} P(-1) &= ((-1)^2 - 2(-1) + 2)^{2026} = (1 + 2 + 2)^{2026} = 5^{2026} \\ \Rightarrow S &= \frac{5^{2026} + 1}{2} \end{aligned}$$

Hence, option (A) is correct.

Question 27 - Solution

The sequence is $a_1 = 3, a_{n+1} = a_n^2 - 2$. Evaluate modulo 13: $a_1 = 3$

$$a_2 = 3^2 - 2 \implies 7$$

$$a_3 = 7^2 - 2 = 47 \implies 8 \pmod{13}$$

$$a_4 = 8^2 - 2 = 62 \implies 10 \pmod{13}$$

$$a_5 = 10^2 - 2 = 98 \implies 7 \pmod{13}$$

The sequence mod 13 follows the cycle 7, 8, 10 repeating every 3 indices for $n \geq 2$. Since $2026 = 3 \times 675 + 1$, the index 2026 acts identically to a_4 within the cycle loop (since $4 \equiv 1 \pmod{3}$) $\implies$ the term resolves to 10.

Hence, option (D) is correct.

Question 28 - Solution

Evaluate $\sum_{k=1}^{\infty} \frac{1}{k(k+1)(k+2)}$. We use partial fractions to telescope the sequence:

$$\frac{1}{k(k+1)(k+2)} = \frac{1}{2} \left(\frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)} \right)$$

The sum becomes a telescoping series evaluating to the first term $k = 1$ scaled by $1/2$:

$$\text{Sum} = \frac{1}{2} \left(\frac{1}{1(2)} - 0 \right) = \frac{1}{4}$$

Hence, option (B) is correct.

Question 29 - Solution

For the expression $\sqrt{x^2 + (y-1)^2} + \sqrt{(x-1)^2 + y^2}$, this represents the geometric sum of Euclidean distances from any point (x, y) to $(0, 1)$ and to $(1, 0)$. The minimum possible total distance is a straight line drawn directly between the two target points $(0, 1)$ and $(1, 0)$. This path crosses through any optimally placed (x, y) . The minimum length is:

$$\sqrt{1^2 + 1^2} = \sqrt{2}$$

Hence, option (D) is correct.

Question 30 - Solution

Evaluate sequence $a_1 = 1, a_{n+1} = a_n + 1/a_n$ for limit $a_n/\sqrt{n}$. Squaring both sides recursively gives:

$$a_{n+1}^2 = a_n^2 + 2 + \frac{1}{a_n^2}$$

Since $a_n \rightarrow \infty$, the $1/a_n^2$ term goes to 0. Thus:

$$a_{n+1}^2 - a_n^2 \approx 2 \implies a_n^2 \approx 2n \implies a_n \approx \sqrt{2n} \implies \frac{a_n}{\sqrt{n}} \rightarrow \sqrt{2}$$

Hence, option (D) is correct.